Arbitrarily long lists of compatible critical relations

Conjecture 00000007762

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For $f_c(z) = z^2 + c$, the official relation polynomials satisfy

$$R_0(c) = 0, \qquad R_{n+1}(c) = R_n(c)^2 + c.$$

We use "nontrivial relation" to mean a nonzero relation polynomial, and compatibility to mean existence of a common parameter solving both equations. The Sylvester resultant certifies the same compatibility for the polynomials below. The statement does not require distinct primitive periods.

Theorem 1. *For every N, the polynomials $R_1, \ldots, R_N$ are distinct and nonzero and have the common root $c = 0$. Their pairwise Sylvester resultants vanish. In particular the asserted bound of three pairwise-compatible nontrivial critical relations fails.*

Proof. Induction gives $R_n(0) = 0$ for all n. Put $b_n = R_n(1)$. Then $b_0 = 0$ and $b_{n+1} = b_n^2 + 1 > b_n$, since $x^2 + 1 - x = (x - 1/2)^2 + 3/4 > 0$. Thus the values b_n are strictly increasing, proving that the polynomials are distinct; $b_n > 0$ for $n \geq 1$ proves that they are nonzero. Since each has zero constant coefficient and is nonzero, each has positive degree.

For any pair, the last column of its Sylvester matrix is zero: only a constant coefficient or an out-of-range zero can occur there. Hence its determinant, the resultant, is zero. Taking the indices $1, 2, 3, 4$ already contradicts the depth-three bound as the problem describes it. The construction in fact gives every finite length. □

Lean correspondence. `R` is a polynomial over the real field, not a list of asserted numerical values. `orbit_correspondence` proves equality with the actual iterates of f_c at zero. Strict increase at $c = 1$, injectivity, nontriviality, positive degree, the shared root, and vanishing of the actual Sylvester determinant are all proved. `ValidChain` restricts indices to these very critical-orbit polynomials and includes all these compatibility requirements. `arbitrary_length` and `no_depth_three_bound` provide the counterexample.

Interpretation boundary. These equations are distinct but algebraically dependent at the fixed parameter $c = 0$. A problem requiring independent equations, exact primitive periods or new irreducible Gleason factors would be different. The final asserted equivalence with a factor statement is not needed to disprove its explicit depth-three conjunct. The text also says the depth is simultaneously unbounded and at most three; the proof above supplies actual orbit polynomials rather than relying only on that verbal contradiction.

Verification. Lean 4.19.0 with Mathlib pinned to `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The accompanying project and verification transcript give the exact theorem and axiom audit. No proof placeholders, native evaluator proof bypass, or custom axiom declarations are used. This is a prepared submission; official acceptance has not been claimed.